HMCS *Canada*/*Queen of Nassau* Archaeological and Structural Assessment
HMCS *Canada* Expedition - 2025

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Abstract
Imagery collected and 3D models generated by the HMCS *Canada* Expedition allowed the National Oceanic and Atmospheric Administration's (NOAA) Office of National Marine Sanctuaries (ONMS) to revisit archaeological research at the HMCS *Canada*/*Queen of Nassau* shipwreck conducted between 2001 and 2005. ONMS found that the rigorous research conducted in the early 2000s established a thorough characterization of the shipwreck. Enhanced data collection methodologies employed by the 2025 research team provided further characterization and interpretation information. Data comparison between the earlier and most recent investigations found the shipwreck to be structurally stable, but that ongoing anthropogenic influences were degrading the archaeological site.

Previous Archaeological Research
Under the direction of archaeologist Tane Casserley, ONMS engaged in archaeological investigations of a deep-water shipwreck offshore of Alligator Reef Lighthouse during three consecutive seasons from 2001-2003. The research established the shipwreck's identity as HMCS *Canada*/*Queen of Nassau*. Research data collected included standard-definition videography, measurements, drawings, and diagnostic artifacts. Casserley processed imagery and measurement data into profile and plan view photomosaics and drawn profile and plan view site maps. These data and products established a baseline site characterization. Casserley documented his historical research and archaeological investigations in his Masters of Arts thesis submitted to East Carolina University's Program in Maritime studies (Casserley 2005). Casserley led the last ONMS site monitoring visit to the shipwreck in 2005. Since that time, visits to the site by technical scuba divers has increased with trips being offered by Key Largo dive charter companies.

Site Interpretation and Structural Analysis
Previous archaeological research established a compelling timeline for the HMCS *Canada*/*Queen of Nassau* wrecking process that significantly augmented the reported story of its loss. Historical reports of the vessel's sinking describe slow water intrusion that intensified into rapid flooding ultimately causing a boiler explosion as the steamship sank stern first (Miami Daily News 1926). Previous investigations did not find evidence of a boiler explosion rupturing the vessel's hull. However, ONMS investigators identified the loss of the starboard propeller and propeller shaft as the likely source of the flooding. The 2025 expedition's high-resolution imaging added additional clarity to the steamship's final moments. Three-dimensional modeling of HMCS *Canada*/*Queen of Nassau*'s stern more fully revealed the scope of the steamship's impact with the seafloor and the near vertical orientation of the hull when it struck the seabed. The impact crushed the cruiser-style stern such that it lost its defining shape. The rudder is no longer extant on the stern (Figure 1).

Figure 1. 3D model capture of HMCS *Canada*/*Queen of Nassau*'s stern (HMCS *Canada* Expedition).

For the first time, the project's complete photogrammetric modeling of the steamship's hull provided observable evidence of the boiler explosion reported by the crew. Review of the 2025 expedition's 3D model by its creator Roger Lacasse, revealed fractures of the hull plating on both starboard and port sides (Figures 2-7). The fractures extended through one course of hull plating down from the gunwale. The fractures were outboard of the boiler room clearly indicated by the position of the smoke stack (Figure 8). The hull fractures may have been further exacerbated by the hull's vertical orientation as it sank. Investigators do not believe the fractures represent damage resulting from the unsupported weight of the bow as it currently rests. No obvious fresh corrosion was apparent in the 2025 imagery suggesting ongoing structural deformation in this area.

Figure 2. Port fracture depicted in a capture from the 3D model (Roger Lacasse, HMCS *Canada* Expedition).

Figure 3. Photograph of the portside fracture (Roger Lacasse, HMCS *Canada* Expedition).

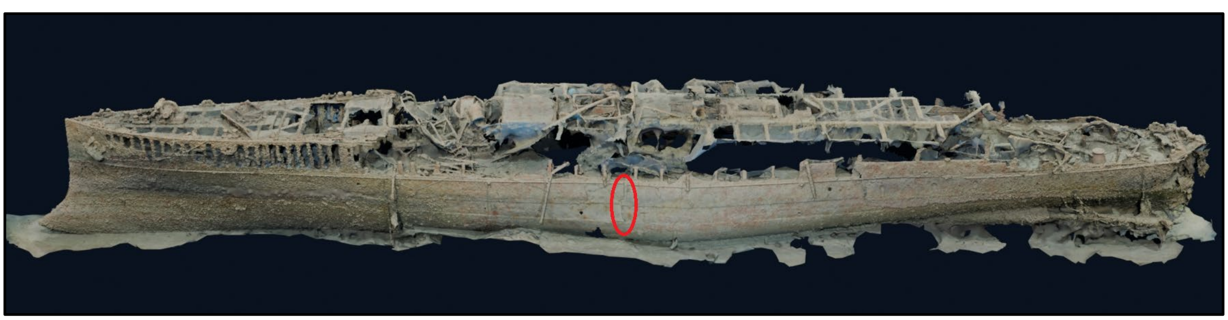

Figure 4. Location of the portside fracture, circled in red, on the 3D model (Roger Lacasse, HMCS *Canada* Expedition).

Figure 5. Starboardside fracture depicted in the 3D model (Roger Lacasse, HMCS *Canada* Expedition).

Figure 6. Starboard fracture photograph, note the empty rivet holes where the plating was pushed outward (Roger Lacasse, HMCS *Canada* Expedition).

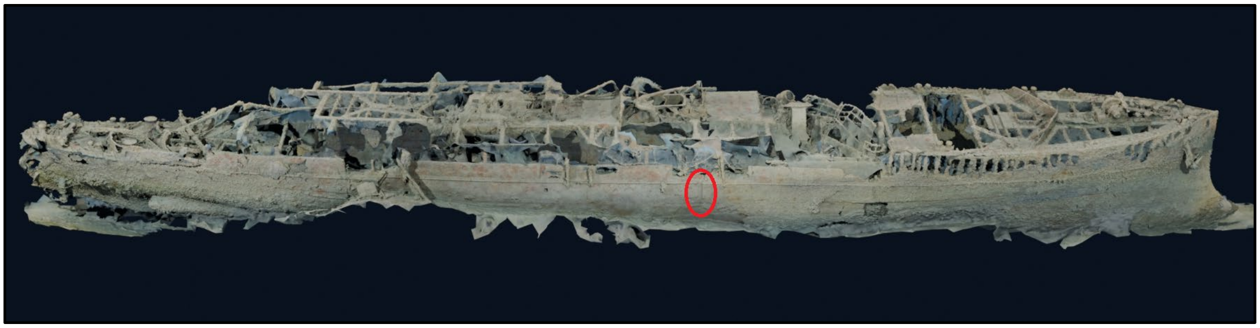

Figure 7. Location of the Starboardside Fracture, Circled in Red, on the 3D Model (Roger Lacasse, HMCS *Canada* Expedition).

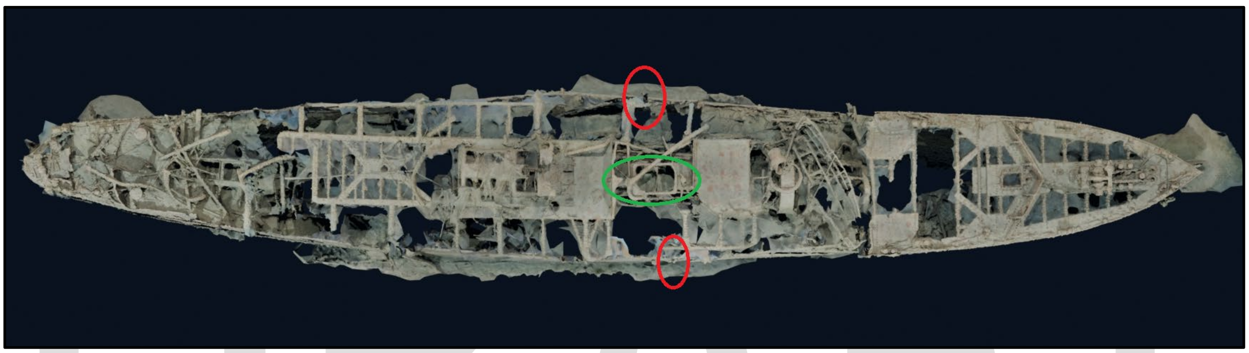

Figure 8. Port and Starboard Fractures (red ovals) in relation to the Smokestack Location (green oval) (Roger Lacasse, HMCS *Canada* Expedition).

Site Change

Comparison of photomosaics/3D models and site imagery from previous ONMS investigations and the HMCS *Canada* Expedition revealed changes to the shipwreck (Figures 9-10). Superstructure deterioration is visible when comparing the baseline ONMS photomosaics to the 2025 3D model plan view. Compared to the baseline planview photomosaic created by Casserley, the bridge railing has shifted further into the vessel's hull. At the stern, the aftermost steel beam that held the weatherdeck cover was pulled down following the baseline investigation and the 2025 expedition. The thin steel plating that surrounded the funnel, showing signs of deterioration in the baseline photomosaic, has completely disappeared in the 2025 expedition imagery and 3D model.

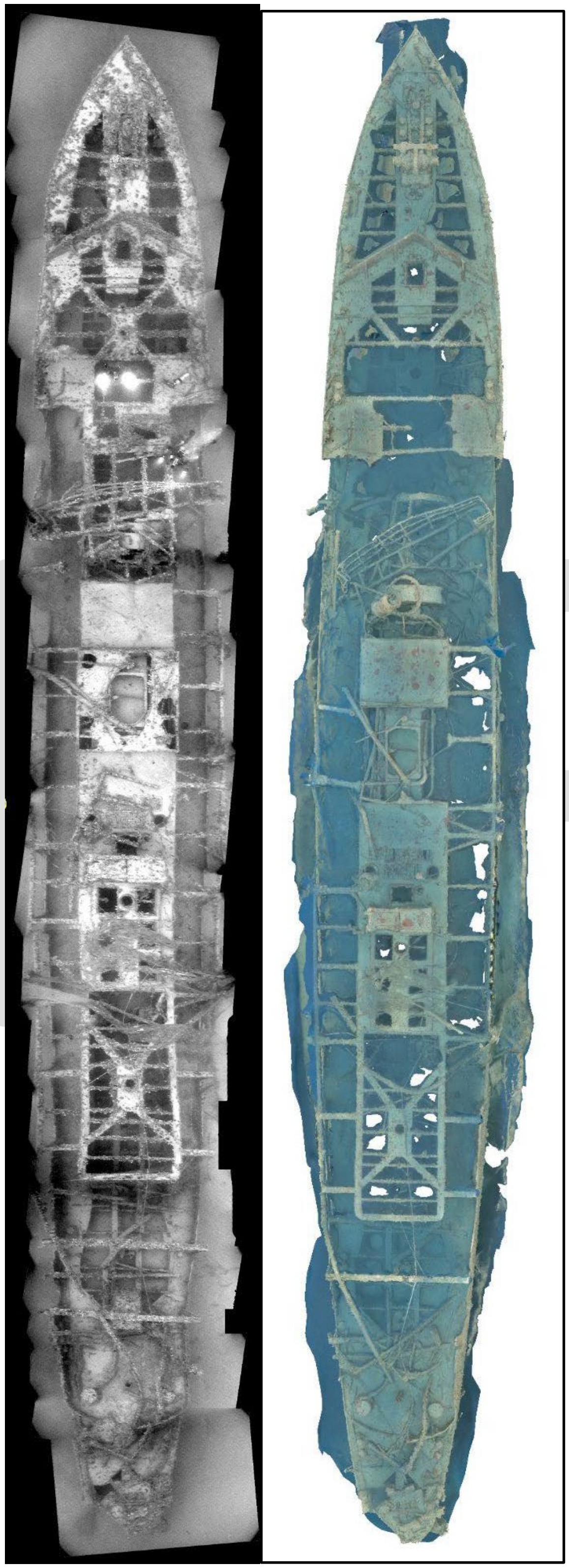

Figure 9. Baseline photomosaic (left) from the 2003 ONMS investigation facilitated comparisons to the 2025 3D model (right) (Tane Casserley, NOAA/ONMS and HMCS *Canada* Expedition).

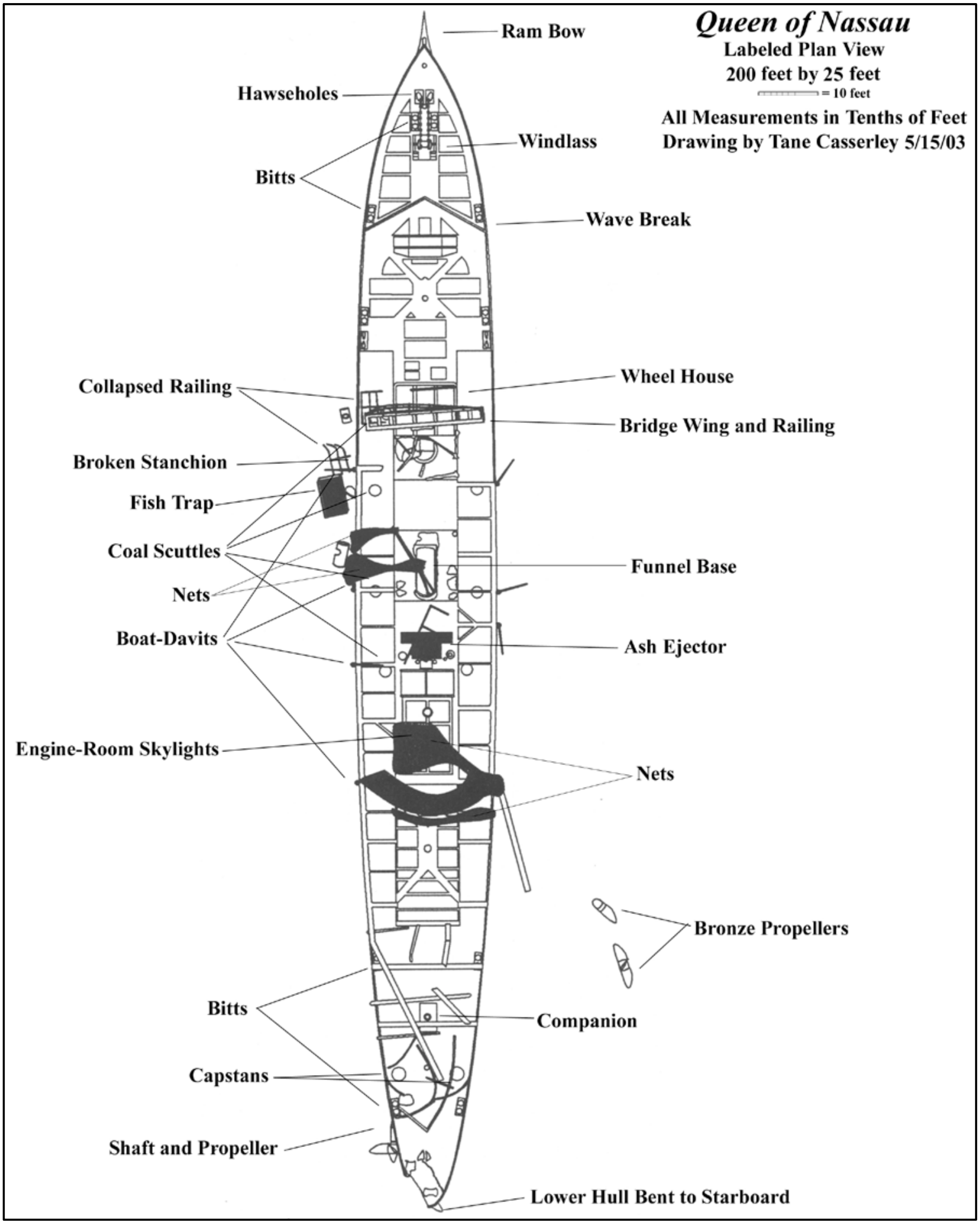

Figure 10. Planview site drawing developed by Casserley from 2003 fieldwork that assisted with determining site changes.

Hull plating that was added to create the steamship's raised top-gallant forecastle appeared to have accelerated deterioration when compared to baseline documentation. This structure was added by the Canadian government in 1912 to improve its seaworthiness (Casserley 2005). The 2003 drawn site map shows little corrosion at the bow (Figure 11). The areas of heightened corrosion documented in 2025 appeared to be localized within the forecastle addition (Figure 12). This change suggested that the steel used in the top-gallant forecastle construction was not as corrosion resistant. Another possibility was that the area received higher levels of oxygenation or held dissimilar metals preferentially corroding the steel hull. A combination of all these factors and others may also be the cause.

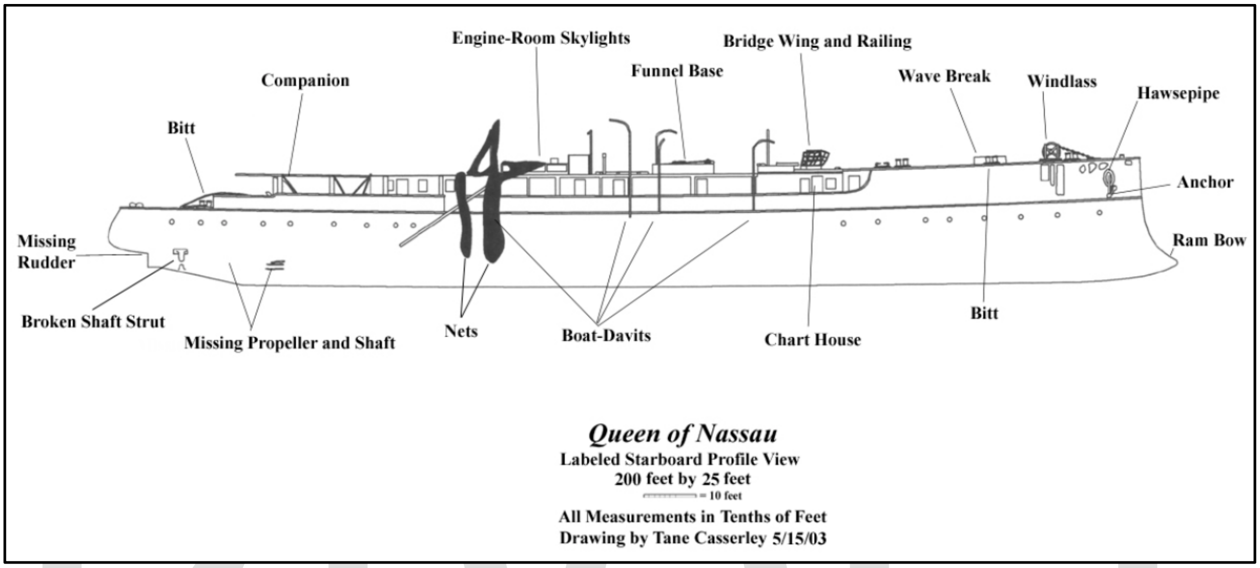

Figure 11. Starboard profile drawing of HMCS *Canada*/*Queen of Nassau* from 2003 fieldwork (Casserley, NOAA ONMS).

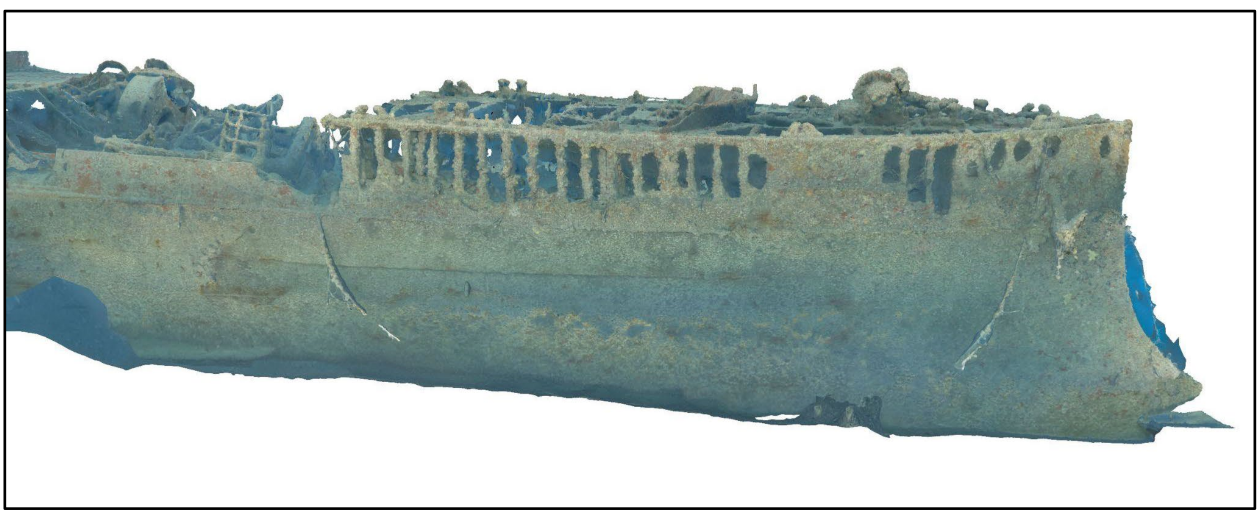

Figure 12. Starboard bow profile view of the 2025 3D model depicting corrosion of the forecastle plating (HMCS *Canada* Expedition).

Anthropogenic Impacts

Imagery collected by the 2025 expedition revealed continued entanglement of the shipwreck in lines and recreational fishing gear. This fouling was noted during the early 2000s ONMS dives, but the quantity of fouling lines was observed to be much greater as more entanglement has occurred and synthetic materials used in fishing line and anchor line does not degrade. Lost anchors and anchor rode have proliferated on the shipwreck. While the fouling fishing line would not have likely caused displacement and damage to weather deck plating and covering beams, the remaining snagged anchor attest to the likely multitude of times anchors hooked the wreck and were then forcibly wrenched free causing some of the structural differences visible when comparing the 2003 imagery with 2025 imagery. Snagged anchors and anchor rode have an additionally pernicious effect on wreck structure. Composed of galvanized steel, stainless steel, and occasionally aluminum, these materials increase corrosion of the HMCS *Canada*/*Queen of Nassau*'s steel hull through galvanic corrosion.

One distinct documented change was in the orientation of two spare propellers that lie on the seafloor off the starboard stern. Figure 13 presents photographs of the propellers taken from nearly the same orientation in 2005 and 2025. The 2005 photo shows the propellers to be much more buried in sediment and at an angle suggesting they plunged into the sediment after sliding off the hull when it sank. Images from the 2025 expedition show the propellers flattened to the sediment and partly entangled with an anchor and anchor rode (Figure 14). Researchers believe that the 2025 photos are clear evidence of disruption by anchoring attempts to access the site by either divers or fishers.

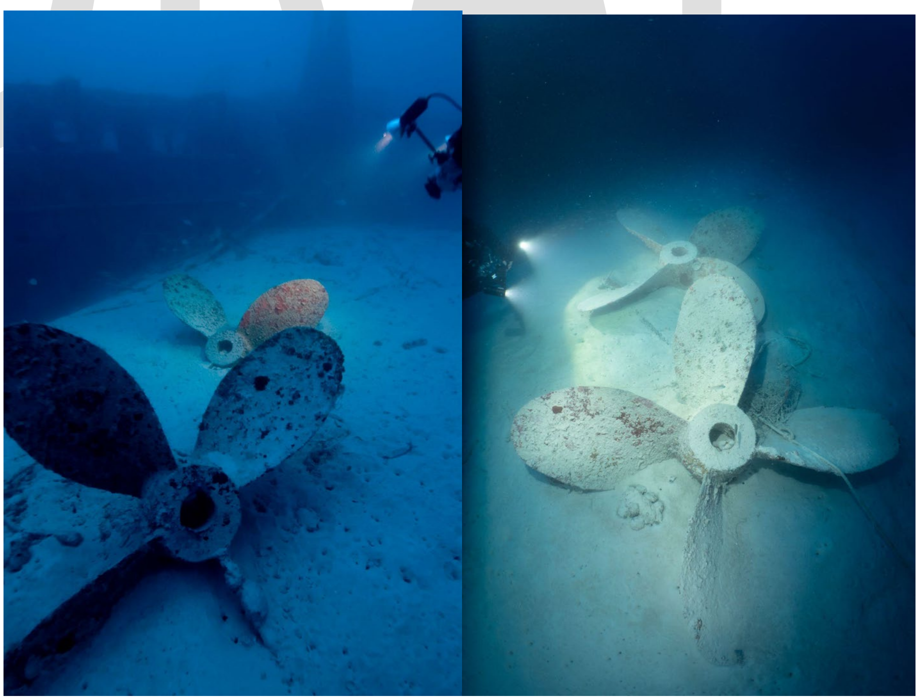

Figure 13. Spare propellers photographed in 2005, left, and 2025, right. (Images courtesy of Tane Casserley, NOAA/ONMS and Ewan Anderson, HMCS *Canada* Expedition 2025).

Figure 14. Anchor entangled with one of HMCS *Canada*/*Queen of Nassau*'s spare propellers (Jason Cook, HMCS *Canada* Expedition)

Methodology Assessment

Structure-from-motion photogrammetric documentation with automated software processing is a relatively new methodology for archaeological site documentation (McCarthy et al. 2019). The simple concept of collecting overlapping imagery of a site from all angles, becomes much more complicated when applied to a 200-foot long largely intact shipwreck only accessible by technical scuba diving. The HMCS *Canada* Expedition team masterfully overcame the challenging environmental conditions in which HMCS *Canada*/*Queen of Nassau* rests to capture the requisite imagery for 3D model creation. Post fieldwork, 3D model creation relies on the employment of specialized software. In general, underwater archaeological site photogrammetry practitioners employ two different software programs, Agisoft Metashape and 3D Flow Zephyr, to generate 3D models. Roger Lacasse, photogrammetry lead for the HMCS *Canada* Expedition, initially used Zephyr to generate the first HMCS *Canada*/*Queen of Nassau* model (Figure 15 black background). Lacasse experienced challenges with the software's representation of HMCS *Canada*/*Queen of Nassau*. He estimated that 90% of the images aligned, but that portions of the hull did not properly render. Lacasse then split the project imagery into seven parts, rendering each part separately in Zephyr. Lacasse then used a second software program, Blender, to align all seven sections into a single model.

Using the same data, ONMS archaeologist Matthew Lawrence used Agisoft Metashape to also generate a 3D model (Figure 15 white background). Lawrence used Metashape presets in the mid range of options to produce the model. Photographic alignment reached 97% of the 10,600 images loaded into the software. The resulting model was a more complete site representation.

The following image presents both models in their planview orientation side by side. Both models do a remarkable job conveying the current condition of HMCS *Canada*/*Queen of Nassau*. There is relatively little difference between the models for the forward three-quarters of the ship except that Metashape depicts imagery from inside the vessel's hull to a greater extent

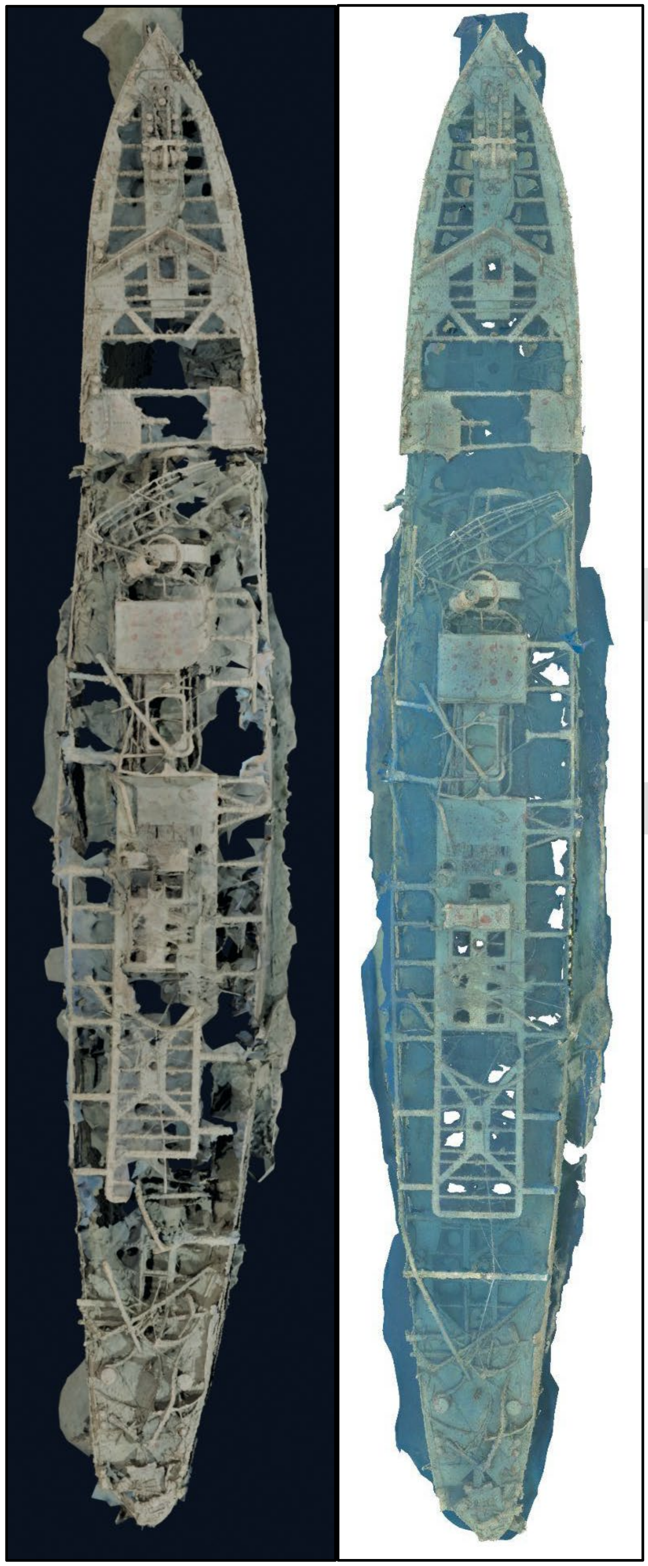

Figure 15. Results from the two different photogrammetric processing methodologies employed by the project. The left image with black background was created using 3D Flow Zephyr/Blender software. The right image with the white background was created using Agisoft Metashape.

than the Zephyr/Blender workflow. The most obvious differences between the models are visible in the stern area. Zephyr/Blender did not accurately generate the weather deck covering structure, truncating or misrepresenting several of the steel beams (Figure 16).

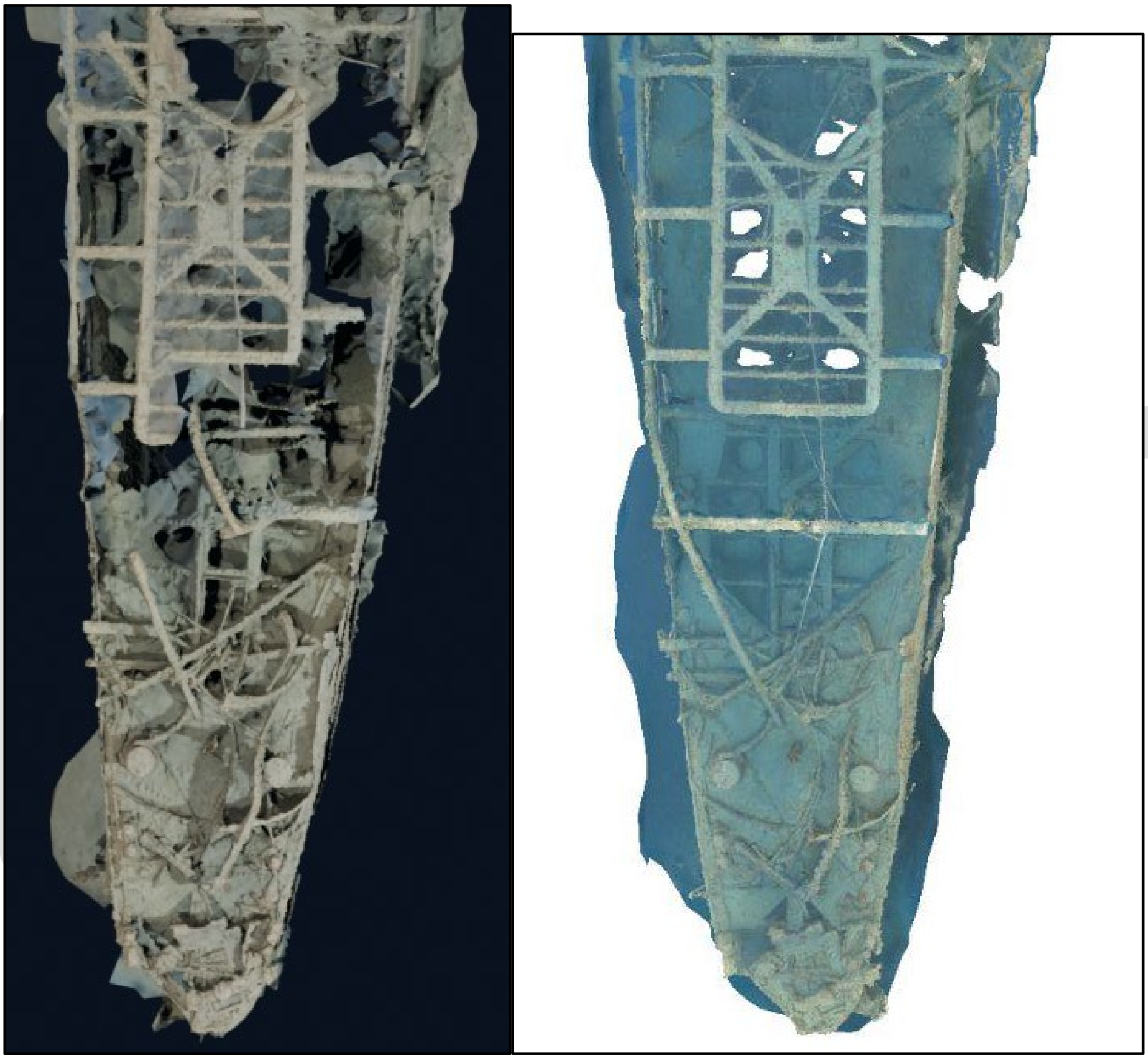

Figure 16. Planview models created with differing software processing methodologies (Zephyr/Blender, left and Metashape, right) show the most distinct differences in the stern section.

Conclusions

HMCS *Canada*/*Queen of Nassau* is a remarkable shipwreck. It encapsulates a period of rapid changes in naval engineering during a very short time period. Its history links Canada and the United States in global trends that dominated the twentieth century, world-wide conflict and tourism. The initial archaeological investigation of the shipwreck brought much of this story to light, but it had lain dormant since. The development of structure-from-motion photogrammetry and technical diving techniques employed by the HMCS *Canada* Expedition was able to reveal the shipwreck to a much greater extent than previous investigations. Comparison against baseline information found that expected archaeological degradation has occurred, but that anthropogenic impacts are accumulating and threatening future generations' enjoyment of the shipwreck. The HMCS *Canada* Expedition Team raised this important heritage back into the collective consciousness and has made it accessible to all.

References Cited

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